

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding Reasoning Paths :

1. **Anatomical snuffbox and ulnar artery injury**: The anatomical snuffbox is located between the extensor pollicis longus and brevis tendons . Injury in this region can compress or damage the radial artery , and the ulnar artery runs superficially near the medial aspect of the wrist , making it susceptible to injury in this area .

2. **Vascular anatomy of the hand**: The ulnar artery is a major artery of the forearm that terminates in the palmar arch , making it a critical source of blood supply to the hand . Injury to this artery can cause significant bleeding .

3. **Clinical presentation of penetrating trauma**: A sharp stab wound to the anatomical snuffbox could directly damage the radial or ulnar artery due to their proximity to the skin and superficial position .

4. **Severe bleeding and vascular involvement**: The excessive bleeding suggests damage to a major vascular structure , with the ulnar artery being a likely candidate given its location and function .

Reasoning Process :

1. **Location of injury**: The anatomical snuffbox is a triangular depression on the lateral side of the wrist , defined by the tendons of the extensor pollicis longus and brevis . The radial artery runs along the radial side of the wrist , while the ulnar artery runs superficially on the medial side . A penetrating wound in this area could easily damage either artery .

2. **Ulnar artery anatomy**: The ulnar artery is a terminal branch of the brachial artery and is the primary blood supply to the hand . It terminates by forming the superficial palmar arch , which provides blood to the ulnar side of the hand . Damage to this artery would result in severe bleeding from the injured hand .

3. **Radial artery anatomy**: The radial artery also runs through the wrist but is more medial than the ulnar artery . It is less likely to be injured in the anatomical snuffbox due to its anatomical position .

4. **Clinical evidence of ulnar artery injury**: The patient presents with excessive bleeding , which aligns with damage to a major artery . The ulnar artery is particularly vulnerable in this scenario because of its superficial location in the wrist area .

5. **Exclusion of other causes**: The bleeding is unlikely to be caused by superficial veins or other smaller arteries , as the described injury suggests a penetrating wound to a major vascular structure . The ulnar artery is the most likely vessel involved due to its size , superficial position , and proximity to the injury site .

Conclusion :

The ulnar artery is the most likely artery damaged in this case , leading to excessive bleeding .

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<answer>

Ulnar artery

</answer>

